Sec: OSR.IIT_*COSC(MODEL-A&APEX) | GTA-5(P2) | Date: 12-04-26
Time: 3 Hrs | 2021_P2 | Max. Marks: 180

KEY SHEET

MATHEMATICS

1	**BC**	2	**ABCD**	3	**ACD**	4	**AC**	5	**AB**
6	**ABD**	7	**4**	8	**40**	9	**70**	10	**16**
11	**9**	12	**45**	13	**C**	14	**B**	15	**B**
16	**D**	17	**8**	18	**1**	19	**6**		

PHYSICS

20	**BC**	21	**CD**	22	**B**	23	**BCD**	24	**AC**
25	**AD**	26	**20**	27	**30**	28	**4**	29	**1**
30	**19.20**	31	**0.80**	32	**C**	33	**D**	34	**B**
35	**C**	36	**4**	37	**1**	38	**6**		

CHEMISTRY

39	**C**	40	**ABC**	41	**ABCD**	42	**ABD**	43	**ABCD**
44	**BCD**	45	**1.50**	46	**182**	47	**5.60**	48	**1.68**
49	**46**	50	**1000**	51	**D**	52	**C**	53	**C**
54	**B**	55	**9**	56	**5**	57	**4**		

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SOLUTIONS
MATHS

1.

Denominator simplifies to $(x^2+4x+2)^2$ and after solving the integral we get

$f(x)=\frac{1}{2}\ln|x^2+4x+2|$

2.

$f'(x)=\frac{1}{\cos x}-\frac{1}{\sin x}=\frac{\sin x-\cos x}{\sin x\cos x}$

$f'(x)=0\Rightarrow x=\frac{\pi}{4}$

$f'\left(\frac{\pi^-}{4}\right)<=0, f'\left(\frac{\pi^+}{4}\right)>0\Rightarrow x=\frac{\pi}{4}$ is a point of minima

$f\left(\frac{\pi}{4}\right)=\int_0^{\frac{\pi}{4}}\sec\theta\,d\theta+\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\cos ec\theta d\theta=2\int_0^{\frac{\pi}{4}}\sec\theta\,d\theta=2\ln\left(1+\sqrt{2}\right)$

$=\ln\left(3+\sqrt{8}\right)$

$a=3,b=8\Rightarrow a+b=11$

3.

$z_1+z_2=-\frac{b}{a}\Rightarrow|z_1+z_2|=1$

$z_1z_2=\frac{c}{a}\Rightarrow|z_1z_2|=1$

Now $|z_1|=1\Rightarrow|z_2|=1$

$|z_1+z_2|\Rightarrow(z_1+z_2)+(\bar{z}_1+\bar{z}_2)=1\Rightarrow 1+1+z_1\bar{z}_2+z_2\bar{z}_1=1\Rightarrow 1+\frac{z_1}{z_2}+\frac{z_2}{z_1}=0$

$\Rightarrow z_1^2+z_2^2-z_1z_2\Rightarrow(z_1+z_2)^2\Rightarrow\left(-\frac{b}{a}\right)^2=\frac{c}{a}\Rightarrow b^2=ac, c^2=ab$

$\therefore$ *Multiply* $a^2=bc$

$\therefore$ *Adding* $\Rightarrow a^2+b^2+c^2=ab+bc+ca\Rightarrow(a-b)^2+(b-c)^2+(c-a)^2=0$

Also $(a-b)^2+(b-c)^2+(c-a)^2=0$

Add & Subtract $2(a-b)(b-c)$

$\Rightarrow(a-b+b-c)+(c-a)^2=2(a-b)(b-c)\Rightarrow|c-a|^2=|a-b||b-c|$

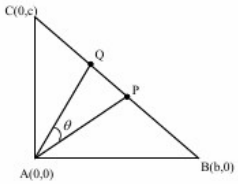
ie by symmetric

$\gamma^2=\alpha\beta,\beta^2=\gamma\alpha,\gamma^2=\alpha\beta\quad\Rightarrow\alpha^2+\beta^2+\gamma^2=\alpha\beta+\beta\gamma+\gamma\alpha$

$\Rightarrow(\alpha-\beta)^2+(\beta-\gamma)^2+(\gamma-\alpha)^2=0\Rightarrow\alpha=\beta=\gamma\Rightarrow|a-b|=|b-c|=|c-a|$

4.

$P=\left(\frac{n+1}{2}b,\frac{n-1}{2}c\right)$



$Q=\left(\frac{n-1}{2}b,\frac{n+1}{2}c\right)\quad m_1=\overline{AP}\text{ slope}=\frac{c(n-1)}{b(n+1)}$

$m_2=\overline{AQ}\text{ slope}=\frac{c}{b}\left(\frac{n+1}{n-1}\right)\therefore\tan\theta=\frac{m_2-m_1}{1+m_1m_2}=\frac{4nbc}{(n^2-1)(b^2+c^2)}$

$=\frac{4nbc}{(n^2-1)a^2}=\frac{4nh}{(n^2-1)a}\quad\left(\because h=\frac{bc}{a}\right)$

$\therefore p=4,q=1,r=1$

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5.

Let $P(x,y,z)$ $(u,2,-3)$ lies on (1)

$(-1,v,3)$ lies on (2)

$$\therefore \quad \frac{1-u}{-1-u}=\frac{-2-2}{v-2}=t \Rightarrow v=2+\frac{4+4u}{1-u} \quad \therefore \quad \left.\begin{matrix} x=(-u-1)t+u \\ y=\frac{(4+4u)}{1-u}t+2 \end{matrix}\right\}$$

$z=6t-3$ (or) $\begin{vmatrix} 0 & z+3 & 3-y \\ 3-z & 0 & x+1 \\ y+2 & 1-x & 0 \end{vmatrix}=0 \Rightarrow 3xy+yz+2zx+6=0$

6.

Possible combinations are (9,9,9,8,8,8,8) / (9,9,9,9,8,7,7) / (9,9,9,9,9,7,7) / (9,9,9,9,9,6,8) / (9,9,9,9,9,9,5)

Required Probability = 4/21

7&8:

$$(\cos x-\sin x)\left(2\tan x+\frac{1}{\cos x}\right)+2=0$$

$$(\cos x)(1-\tan x)\left(2\tan x+\frac{1}{\cos x}\right)+2=0$$

$$(1-\tan x)(\cos x)\left(2\tan x+\frac{1}{\cos x}\right)+2=0$$

$$(1-\tan x)(2\tan x\cos x+1)+2=0$$

Consider $\tan x=t$ $\cos x=\frac{1}{\sec x}=\frac{1}{\sqrt{t^2+1}}$

$$(1-t)\left(\frac{2t}{\sqrt{t^2+1}}+1\right)+2=0 \quad \left(\frac{2t}{\sqrt{t^2+1}}+1\right)=\frac{2}{t-1} \quad \left(\frac{2t}{\sqrt{t^2+1}}\right)=\frac{2}{t-1}-1=\frac{3-t}{t-1}$$

Squaring on both sides, we get $\frac{4t^2}{t^2+1}=\frac{9+t^2-6t}{t^2+1-2t}$

Simplifying the given equation, we get

$3t^4-2t^3-6t^2+6t-9=0$

Rearranging the given equation, we get

$3(t^4-9)-2t(t^2-3)-6(t^2-3)=0$

$(t^2-3)(3t^2-2t+3)=0 \quad t=\sqrt{3},-\sqrt{3}$

$t=\sqrt{3}$ is the only solution possible

9 & 10:

$$g(x)=x^2f(1)+xf'(x)+f''(x)$$

$$f(x)=x^2+ax+b, \quad a=g'(1), \quad b=g''(2)$$

$$g(x)=(a+b+1)x^2+x(2x+a)+2$$

$$=(a+b+3)x^2+ax+2$$

$$g'(x)=(a+b+3)(2x)+a$$

$$g'(1)=2(a+b+3)+a \Rightarrow a+b+3=0$$

$$\Rightarrow g(x)=ax+2$$

$$g''(x)=0 \Rightarrow b=0$$

$$a=-3$$

$$f(x)=x^2-3x, g(x)=-3x+2$$

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11.

$$AB = A^2B^2 - (AB)^2$$

$$A = A^2B - ABA$$

$$A(I - AB + BA) = 0$$

$|A| = 0$

For 2 x 2 matrices A and B the following relation holds good for some scalar x

$$|A + xB| = |A| + x^2|B| + x((TrA)(TrB) - (TrAB))$$

$$|A + 2B| - |B + 2A|$$

$$(|A| + 4|B| + 2((TrA)(TrB) - (TrAB)) - |B| + 4|A| + 2((TrA)(TrB) - (TrAB)))$$

$$= 3|B| - 3|A| = 3(3) - 3(0) = 9$$

$$|A - 4B| - |B - 4A|$$

$$= (|A| + 16|B| - 4((TrA)(TrB) - (TrAB)) + |B| + 16|A| - 4((TrA)(TrB) - (TrAB)))$$

$$= 15|B| - 15|A| = 15(3) - 15(0) = 45$$

13. $\begin{vmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix} = 0$

$$\Rightarrow -(\alpha + \beta + \gamma)(\alpha^2 + \beta^2 + \gamma^2 - \sum \alpha\beta) = 0$$

$$\Rightarrow -(-a)(a^2 - 2b - b) = 0$$

$$\Rightarrow a(a^2 - 3b) = 0$$

$$\Rightarrow a = 3b \Rightarrow \frac{a^2}{b} = 3$$

14. $P_1 : x + 2y - 3z = a$

$P_2 : 2x + 6y - 11z = b$

$P_3 : x - 2y + 7z = c$

Clearly

$5P_1 = 2P_2 + P_3$

If $5a = 2b + c$

$\Rightarrow$ All the planes sharing a line of intersection

$\Rightarrow$ Infinite solutions

15&16:

$$\sum_{i=1}^{18}(x_i - \alpha) = 36, \sum_{i=1}^{18}(x_i - \beta)^2 = 90$$

$$\Rightarrow \sum_{i=1}^{18} x_i = 18(\alpha + 2), \sum_{i=1}^{18} x_i^2 - 2\beta\sum x_i + 18\beta^2 = 90$$

Hence $\sum x_i^2 = 90 - 18\beta^2 + 36\beta(\alpha + 2)$

Given $\frac{\sum x_i^2}{18} - \left(\frac{\sum x_i}{18}\right)^2 = 1$

$$\Rightarrow 90 - 18\beta^2 + 36\beta(\alpha + 2) - 18(\alpha + 2)^2 = 18$$

$$\Rightarrow 5 - \beta^2 + 2\alpha\beta + 4\beta - \alpha^2 - 4\alpha - 4 = 1$$

$$\Rightarrow (\alpha - \beta)^2 + 4(\alpha - \beta) = 0 \Rightarrow |\alpha - \beta| = 0 \text{ or } 4$$

As α amd β are distinct $|\alpha - \beta| = 4$

$A = (-\infty, 1) \cup (3, \infty)$

$B = (-\infty, -2) \cup (2, \infty)$

$C = (-\infty, 2] \cup [6, \infty)$

$So, A \cap B \cap C = (-\infty, -2) \cup [6, \infty)$

$z \cap (A \cap B \cap C)' = \{-2, -1, 0, 1, 2, 3, 4, 5\}$

Hence no. of its subsets $= 2^8 = 256$.

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17. $R.N.W = {}^{10}C_4\ {}^{6}C_6 + {}^{6}C_5 + {}^{6}C_4 + {}^{6}C_3 = 8820$

18. $GE = \sqrt{100+\sqrt{n}} + \sqrt{100-\sqrt[4]{n}} = p(say)$

$p^2 = 200 + 2\sqrt{10000-n}$ in to be maximum for n to be minimum

Also p^2 is integer $\Rightarrow \sqrt{10000-n}$ is integer

But $200 + 2\sqrt{10000} = 400 = 20^2$

$\therefore 200 + 2\sqrt{10000-n} < 20^2$

$\Rightarrow 200 + 2\sqrt{10000-n} = 18^2$ $(\because P \text{ in even})$

$\Rightarrow \sqrt{10000-n} = 62$

$\Rightarrow n = 100^2 - 62^2 \Rightarrow n = (62)(38)$

$\Rightarrow n = 6156$

$\Rightarrow n = 2^2.3^4(19)^1$

19. Distance between circle and parabola

$x^2 + (y-12)^2 = 1 and\ y^2 = 4x$

PHYSICS

20.



$i_{initial} = \dfrac{V_0}{R} \Rightarrow$ constant $\Rightarrow$ P.D. across resistor is

Constant $= V_0$

$q = C_0V_0 - \dfrac{V_0}{R}.t$

$\therefore C = \dfrac{q}{V_0} = C_0\left(1 - \dfrac{t}{RC_0}\right)$

For $q = 0, t = RC$

Heat $= i^2Rt = \left(\dfrac{V_0}{R}\right)^2.R.RC = CV_0^2$

21.

The velocity parallel to the plane is unaltered by the impacts, so that the distance described parallel to the plane will be zero at the end of a time t given by :

$0 = vt\cos(\theta-\alpha) - \dfrac{(g\sin\alpha)t^2}{2}$

$\Rightarrow \quad t = \dfrac{2v\cos(\theta-\alpha)}{g\sin\alpha}$

Also, since the elasticity is perfect, the velocity perpendicular to the plane is just reversed at each impact. The time of flight for each trajectory is thus twice the time in which the velocity $v\sin(\theta-\alpha)$ is destroyed by $g\cos\alpha$, and thus $t = \dfrac{2v\sin(\theta-\alpha)}{g\cos\alpha}$

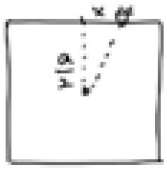
Clearly the particle will return to the point of projection if the first of these is some multiple, n, of the second, i.e. if

$\dfrac{2v\cos(\theta-\alpha)}{g\sin\alpha} = n\dfrac{2v\sin(\theta-\alpha)}{\cos\alpha}$

i.e. if $\cot\alpha\ .\ \cot(\theta-\alpha)$ is an integer.

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22.



$$T = 2\pi\sqrt{\left(\frac{\frac{5ma^2}{12}+mx^2}{mg\sqrt{\frac{a^2}{4}+x^2}}\right)}$$ monotonous

$$I = \frac{ma^2}{6}+\frac{ma^2}{4}+mx^2 \quad \therefore T_{(x=0)} = 2\pi\sqrt{\frac{\frac{5ma^2}{12}}{mg.\frac{a}{2}}} = \sqrt{2}\sec$$

$$T_{\left(x=\frac{a}{2}\right)} = 2\pi\sqrt{\frac{\frac{2}{3}ma^2}{mg\frac{a}{\sqrt{2}}}} \simeq 1.5\sec$$

23. A point with zero field can exits in the YZ plane



The result of the four component fields will be perpendicular to $Y-Z$ plane (if it is not zero)

24. $\lambda_{k\alpha} = 0.71\overset{0}{\text{A}}$ $\quad \frac{1}{\lambda_{k\beta}} = \frac{1}{\lambda_{k\alpha}}+\frac{1}{\lambda L_\alpha}$

$\lambda_{k\beta} = 0.63\overset{o}{\text{A}}$

$\lambda_{cut-off} = 0.4\overset{o}{\text{A}} = \frac{12400eV-\overset{o}{\text{A}}}{E}$ $\quad \lambda_{L\beta} = 5.6\overset{o}{\text{A}}$

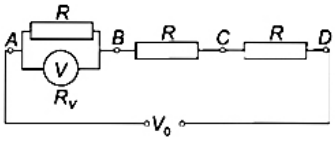
$\therefore eV_a = 31KeV \Rightarrow V_a = 31KV$

If V_a is doubled, λ_C becomes half, $\lambda_{K\alpha}-\lambda_{K\beta}$ remains same

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25.

From the first measurements we know that the three resistances are equal (R).



Let R_V stand for the internal resistance of the voltmeter. Then the potential differences between point pairs AB, BC, CD are divided in the ratio of the resistances measured between these points. (See the figure). From this, the potential difference between point pair AB is

$$V_{AB} = \frac{U_0}{\frac{R \cdot R_V}{R+R_V} + 2R} \cdot \frac{R \cdot R_V}{R+R_V}.$$

After simplifying and reorganising, the parametric equation for the potential difference, which was measured to be 20 V is

$$V_{AB} = \frac{V_0 \cdot R_V}{2R + 3R_V}.$$

If the voltmeter is connected between points A and C, the above mentioned proportional part can be calculated from the following formula:

$$V_{AC} = \frac{V_0}{\frac{2R \cdot R_V}{2R+R_V} + R} \cdot \frac{2R \cdot R_V}{2R+R_V},$$

after simplifying, this surprisingly results in exactly twice of value V_{AC}:

$$V_{AC} = 2 \cdot \frac{V_0 \cdot R_V}{2R - 3R_V} = 2V_{AB} = 40\ V.$$

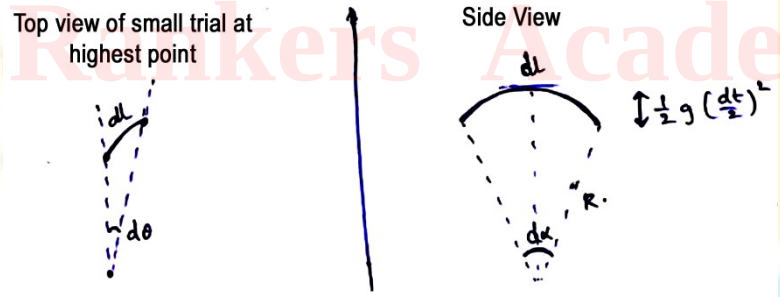
As $R_b \approx 0$,

$$V_{AD} = V_0 = 62\ V,$$

which is not three times V_{AB}!

26. $$R = \frac{(u\cos\theta)^2}{g} = \frac{20^2 \cdot \frac{1}{2}}{10} = 20\ m$$

27.



If sprinkler is not rotating:

$$d\theta = 0,\ dl = U_x dt$$

If sprinkler is rotating:

$$dl = \sqrt{(U_x dt)^2 + \left(\frac{L}{2}.d\theta\right)^2}$$ L:Horizontla Range

$$U_x dt = R.d\alpha \quad \ldots (1)$$

$$\frac{1}{2}g\left(\frac{dt}{2}\right)^2 = R\left(1-\cos\frac{d\alpha}{2}\right) \quad d\theta = \omega dt;\ dl = Rd\alpha$$

$$\frac{1}{2}g\left(\frac{dt}{2}\right)^2 = R\left(1-\cos\frac{d\alpha}{2}\right) \ldots (2)$$

Solving (1) & (2)

Solving (1) & (2)

$$R = \frac{U_x^2 + \left(\frac{\omega L}{2}\right)^2}{g}$$

$$R = \frac{U_x^2}{g}$$

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28. $L=\left(\frac{mL^2}{3}+mR^2+m\left(L+R\right)^2+m\left(L+R\right)^2\right)w$

$=mR^2\left(\frac{1}{3}+1+4+4\right)w=\frac{3}{7}.1.\frac{28}{3}=4\ kg.m^2/s$

29. $a_A=\omega^2R \qquad a_B=\omega^2\left(L+R\right)$

$a_{A/B}=\omega^2L=1\ m/s^2$

30. Energy absorbed = Energy radiated

$=\left(0.6\right)\left(4\pi\left(2R\right)^2\right)2\sigma.T^4$

$=19.2\pi R^2\sigma T^4$

31. Enegy reflected $=\left(0.2\right)\left(4\pi r^2T^4\sigma\right)$

$=0.8\pi r^2\sigma T^4$

32.

Consider a circular orbit of radius R_C. Considering centripetal forces,

$$\frac{GMs}{R_c^2}=\frac{v_c^2}{R_c}$$

Consider an elliptical orbit that has closet approach R_1 and farthest point R_2. Considering energy,

$$\frac{1}{2}v^2-\frac{GMs}{r}=E$$

Considering angular momentum at the closest and farthest points,

$v_1R_1 = v_2R_2$.

Combining, and focusing on the closest point,

$$\frac{1}{2}v_1^2-\frac{GMs}{R_1}=\frac{1}{2}v_1^2\left(\frac{R_1}{R_2}\right)^2-\frac{GMs}{R_2}$$

which can be solved for v_1,

$$\frac{1}{2}v_1^2\left(1-\left(\frac{R_1}{R_2}\right)^2\right)=GMs\frac{R_2-R_1}{R_1R_2}$$

or, if we let $R_1/R_2 = \alpha$,

$$\frac{1}{2}v_1^2\left(1-\alpha^2\right)=\frac{GMs}{R_1}\left(1-\alpha\right)$$

which simplifies into

$$v_1=v_E\sqrt{\frac{2}{1+\alpha}}$$

This is necessarily greater than v_E, so the boost is

$$\Delta v_1=v_E\left(\sqrt{\frac{2}{1+\alpha}}-1\right)$$

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33.

$$\frac{1}{2}v_2^2\left(1-(1/\alpha)^2\right)=\frac{GMs}{R_2}\left(1-(1/\alpha)\right)$$

which simplifies to

$$v_2 = v_M\sqrt{\frac{2}{1+1/\alpha}}$$

T is less than v_M, so the rocket must receive a second positive boost :

$$\Delta v_2 = v_M\left(1-\sqrt{\frac{2}{1+1/\alpha}}\right)$$

Note, however, that we want the answer in terms of v_E. So there is some more math to do. First, by Kepler's law,

$$v_E^2 R_E = v_M^2 R_M,$$

which implies

$$v_M = v_E\sqrt{\alpha}$$

Then

$$\Delta v_2 = v_E\sqrt{\alpha}\left(1-\sqrt{\frac{2}{1+\frac{1}{\alpha}}}\right)$$

34. $W_{gas} = P_0.A(3) + S_1.A.(2)g.5 + S_2.A(1).g.(2.5)$

$= 10^5.10^{-2}.(3) + 10^3.12^{-2}.2.10.5 + 2\times10^3\times10^{-2}\times10.2.5$

$= 3000 + 1000 + 500 = 4500J$

35. $\Delta Q = \Delta U + W$

$= \frac{3}{2}\left(P_f V_f - P_i.V_i\right) + 4500$

$= \frac{3}{2}\left(1.6\times10^5\times10^{-2}\times7 - 1.4\times10^5\times10^{-2}\times4\right) + 4500$

$= \frac{3}{2}(11200-5600) + 4500 = 12900\ J$

36. Let individual intensities at the centre be $9I$ & I

$$I_{max} = \left(\sqrt{9I}+\sqrt{I}\right)^2 = 25W/m^2 \Rightarrow I = \frac{25}{16}W/m^2$$

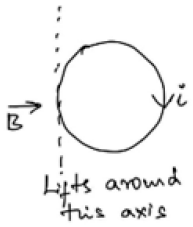
$$\frac{I^1}{I} = \frac{(d)^2}{\left(\frac{d}{\cos\theta}\right)^2} = \frac{16}{25}$$

$$\Delta\phi = \frac{2\pi}{\lambda}.d\sin\theta = \pi$$

$$\therefore I_{min} = \left(\sqrt{9I^1}-\sqrt{I^1}\right)^2 = 4I^1 = 4.\frac{16}{25}.I = 4W/m^2$$

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37.

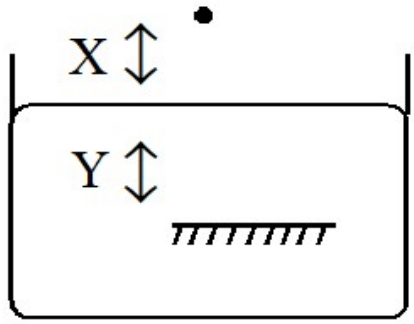


$$i = \frac{\pi r^2 .8t}{R}$$

$$mgr = iAB_x = \frac{\pi r^2 .8t}{R}.\pi r^2 .4$$

$$t = \frac{mgr}{\pi^2 .r^3 .(32)} \simeq 1 \text{ sec}$$

38.



1st refraction; image distance $= \mu x$

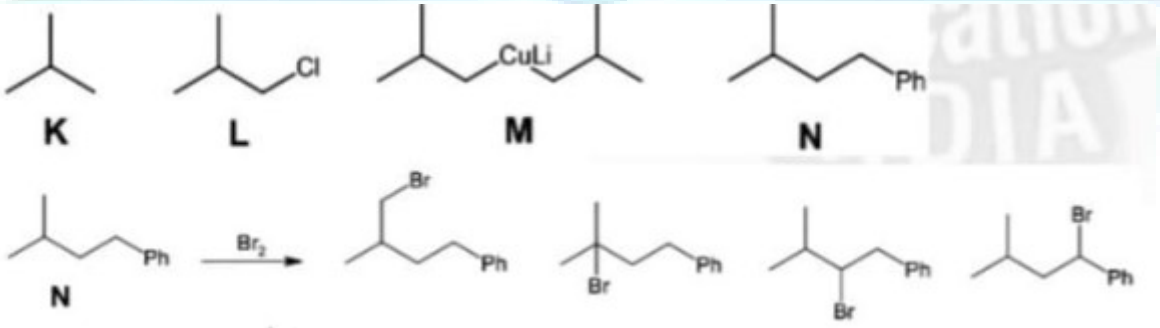
2st reflection; image distance $= \mu x + y$

2nd refraction; image distance $= \frac{\mu x + 2y}{\mu}$ (from surface of water)

$$\therefore V = \frac{d}{dt}\left(x + \frac{2}{\mu}y\right) = 0 + \frac{2}{\left(\frac{4}{5}\right)}.\frac{dy}{dt} = 6 m/s$$

CHEMISTRY

39.



40. A) V is the most basic nitrogen of all the labelled nitrogen atoms

B) III is in conjugation with the ring. I is sp^2 hybridized while II is sp^3 hybridized

C) Self-explanatory

D) V is more basic than I and VII

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$$A(g) \rightarrow 2B(g) + C(g)$$

$t = 0$ 400

$t = 20$ $400 - x$ $2x$ x

So $400 - x + 2x + x = 1000$ $x = 300\, mm\ Hg$

$$K = \frac{1}{20}\ln\frac{400}{100} = \frac{2\ln 2}{20} = 0.0693\,\text{min}^{-1}$$

$$t_{1/2} = \frac{\ln 2}{K} = \frac{0.693}{0.0693} = 10\,\text{min}$$

$$\frac{30}{10} \times \ln 2 = \ln\frac{400}{400 - x}$$

$$8 = \frac{400}{400 - x} \Rightarrow 400 - x = 50$$

$$\Rightarrow x = 350\ mm\ Hg$$

Partial pressure of C after 30 min = 350 mm Hg

41. Total pressure after 30 min = $400 + 2x = 400 + 700 = 1100\ mm\ Hg$

42. Trisilyl amine is weak Lewis base than trimethyl amine due to $p\pi - d\pi$ bonding between N and Si.

43. (C) C – O bond length: Maximum in $\left[V(CO)_6\right]^-$ due to greater extent of synergic bonding

D) $\left[Co(H_2O)_6\right]^{3+} > \left[Co(NH_3)_6\right]^{3+} > \left[Co(CN)_6\right]^{3-}$: is the correct order of absorption of wave length in the visible region (ligand field strength is: $H_2O < NH_3 < CN^-$

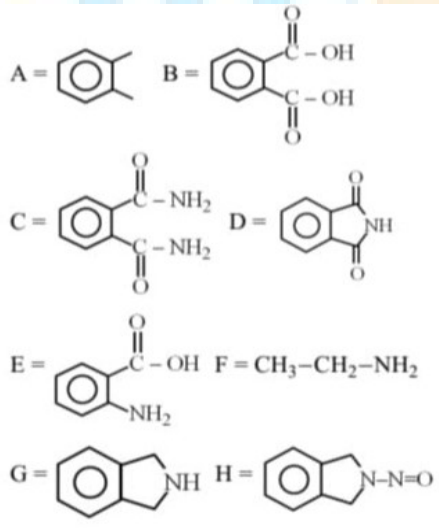
44. $MnO_2 + 4HCl \xrightarrow{\Delta} MnCl_2 + Cl_2 + 2H_2O$

$4FeS_2 + 11O_2 \rightarrow 2Fe_2O_3 + 8SO_2(g)$

$SO_2 + Cl_2 \rightarrow SO_2Cl_2(l)$

$P_4 + 10SO_2Cl_2 \rightarrow 4PCl_5 + 10SO_2$

45 & 46:



DBE of H = 6

DBE of A = 4

∴ ratio of DBE = $\frac{6}{4} = 1.50$

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47. $\Delta S_{sys} = nR\ln\frac{V_2}{V_1} = 8\times 0.7 = 5.6 J/mol\,K$

48. $\Delta S_{sys} = nR\ln\frac{V_2}{V_1} = 8\times 0.7 = 5.6 J/mol\,K$

$\Delta G_{sys} = \Delta H - T\Delta S = 0 - 300\times 5.6\times 10^{-3} = -1.68 kJ/mol;$ $\left(\because \Delta H_{sys} = 0\right)$

49. $IO_3^- + 3HSO_3^- \rightarrow I^- + 3SO_4^{2-} + 3H^+$

198 grams of $NaIO_3$ reacts with $\rightarrow (3\times 104)$ *grams of* $NaHSO_3$

$\therefore (5\times 5.8)$ grams of $NaIO_3$ reacts with $\rightarrow ?$ *grams of* $NaHSO_3$

$$= \frac{3\times 104\times 5\times 5.8}{198} = 45.69 \approx 46 \text{ grams}$$

50. Number of moles of I^- formed in the I step $= \frac{5\times 5.8}{198}$

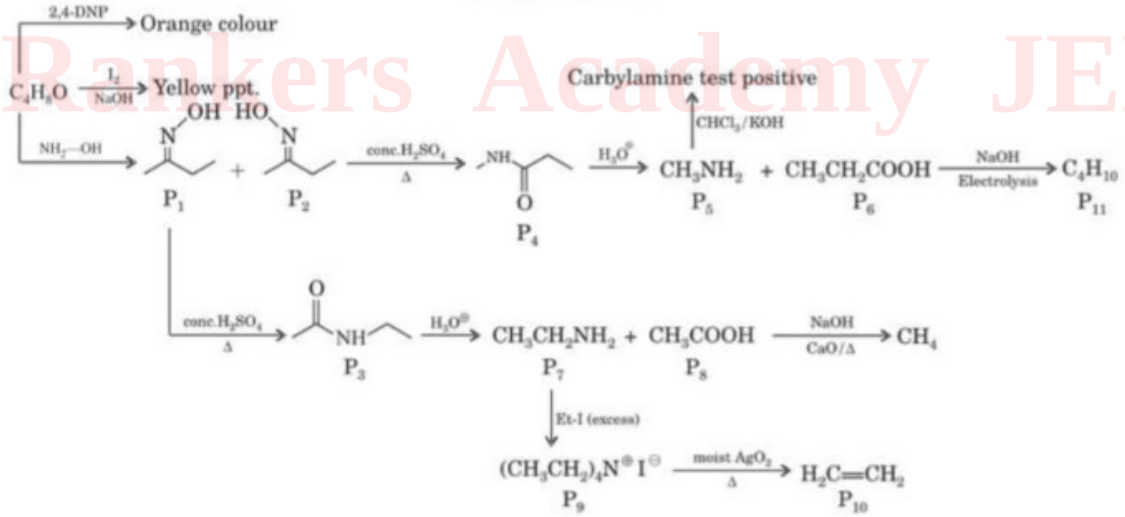
$IO_3^- + 5I^- + 6H^+ \rightarrow 3I_2 + 3H_2O$

5 *moles of* $I^- \equiv 198$ *g of* $NaIO_3$

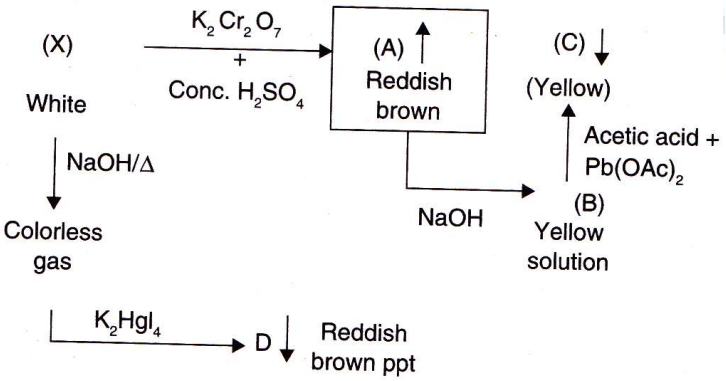
$\therefore \frac{5\times 5.8}{198}$ *moles of* $I^- \equiv ?$ *g of* $NaIO_3 = \frac{198\times 5\times 5.8}{5\times 198} = 5.8$ *grams of* $NaIO_3$

5.8 grams of $NaIO_3$ is present in one litre (1000 ml) solution

51 & 52.



53.



(X) is NH_4Cl

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It liberates CrO_2Cl_2 (A) with $K_2Cr_2O_7$ and conc. H_2SO_4 (Chromyl chloride test)

$CrO_2Cl_2 + 4\ NaOH \rightarrow Na_2CrO_4 + 2\ NaCl + 2H_2O$

$Na_2CrO_4 + Pb^{2+} \rightarrow PbCrO_4 \downarrow (yellow\ ppt) + 2Na^+$

$NH_4Cl + NaOH \xrightarrow{\Delta} NH_3 \uparrow + NaCl + H_2O$

$NH_3 \xrightarrow{K_2HgI_4} I - Hg - O - Hg - NH_2 \downarrow$ (Reddish brown ppt.)

54. $X = NH_4Cl; A = CrO_2Cl_2; B = Na_2CrO_4; C = PbCrO_4$

$NH_4Cl + NaOH \rightarrow NaCl + NH_3 + H_2O$

55. $AgA(s) \underset{}{\overset{3\times10^{-14}}{\rightleftharpoons}} \underset{(s_1+s_2)}{Ag^+(aq)} + \underset{s_1}{A^-(aq)}$

$AgB(s) \underset{}{\overset{1\times10^{-14}}{\rightleftharpoons}} \underset{(s_1+s_2)}{Ag^+(aq)} + \underset{s_2}{B^-(aq)}$

$s_1(s_1+s_2) = 3\times10^{-14} ----\rightarrow (1)$

$s_2(s_1+s_2) = 1\times10^{-14} ----\rightarrow (2)$

Solving, we get; $s_1 = 1.5\times10^{-7}\ M; s_2 = 0.5\times10^{-7}\ M$

$\therefore [A^-] = s_1 = 1.5\times10^{-7}\ M; [B^-] = s_2 = 0.5\times10^{-7}\ M; [Ag^+] = (s_1+s_2) = 2\times10^{-7}\ M$

$\lambda_{[Ag^+]} = \frac{\kappa_{Ag^+}\times1000}{[Ag^+]} \Rightarrow 60 = \frac{\kappa_{Ag^+}\times1000}{2\times10^{-7}} \Rightarrow \kappa_{[Ag^+]} = 120\times10^{-10}\ Scm^{-1}$

$\lambda_{[A^-]} = \frac{\kappa_{A^-}\times1000}{[A^-]} \Rightarrow 80 = \frac{\kappa_{A^-}\times1000}{1.5\times10^{-7}} \Rightarrow \kappa_{[A^-]} = 120\times10^{-10}\ Scm^{-1}$

$\kappa_{Ag^+} + \kappa_{A^-} + \kappa_{B^-} = 375\times10^{-10}$

$(120\times10^{-10}) + (120\times10^{-10}) + \kappa_{B^-} = 375\times10^{-10} \Rightarrow \kappa_{B^-} = (375-240)\times10^{-10} = 135\times10^{-10}$

$\lambda_{[B^-]} = \frac{\kappa_{B^-}\times1000}{[B^-]} = \frac{(135\times10^{-10})\times1000}{0.5\times10^{-7}} = 270\ Scm^2mol^{-1}$

56. $\frac{V_{Ne}}{V_{He}} = \sqrt{\frac{T_{Ne}}{M_{Ne}}\times\frac{M_{He}}{T_{He}}} = \sqrt{\frac{1000}{20}\times\frac{4}{200}} = 1$

$\frac{\lambda_{He}}{\lambda_{Ne}} = \frac{m_{Ne}\times v_{Ne}}{m_{He}\times v_{He}} = \frac{20}{4} = 5$

57. $NH_4NO_2, NaN_3, (NH_4)_2Cr_2O_7, Ba(N_3)_2$ on heating give nitrogen gas.

$NH_4NO_2 \xrightarrow{\Delta} N_2 + 2H_2O$

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